

Chapter 12

Principles of Neutralization Titrations

Fundamentals of Analytical Chemistry, Skoog, et al., 10th Ed., 2022

Chapters 7 & 12 Problems:

7-4, 7-20^q, 7-21^q,

12-13, 12-17, 12-23, 12-25, 12-27^q, 12-29^q, 12-33, 12-41a^q

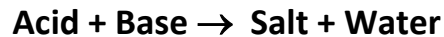
^qrequires quadratic

TABLE D.1 Dissociation Constants for Acids at 25°C

Name	Formula	K_{a1}	K_{a2}	K_{a3}
Acetic	$\text{HC}_2\text{H}_3\text{O}_2$	1.8×10^{-5}		
Arsenic	H_3AsO_4	5.6×10^{-3}	1.0×10^{-7}	3.0×10^{-12}
Arsenous	H_3AsO_3	5.1×10^{-10}		
Ascorbic	$\text{HC}_6\text{H}_7\text{O}_6$	8.0×10^{-5}	1.6×10^{-12}	
Benzoic	$\text{HC}_7\text{H}_5\text{O}_2$	6.3×10^{-5}		
Boric	H_3BO_3	5.8×10^{-10}		
Butanoic	$\text{HC}_4\text{H}_7\text{O}_2$	1.5×10^{-5}		
Carbonic	H_2CO_3	4.3×10^{-7}	5.6×10^{-11}	
Chloroacetic	$\text{HC}_2\text{H}_2\text{O}_2\text{Cl}$	1.4×10^{-3}		
Chlorous	HClO_2	1.1×10^{-2}		
Citric	$\text{H}_3\text{C}_6\text{H}_5\text{O}_7$	7.4×10^{-4}	1.7×10^{-5}	4.0×10^{-7}
Cyanic	HCNO	3.5×10^{-4}		
Formic	HCHO_2	1.8×10^{-4}		
Hydroazoic	HN_3	1.9×10^{-5}		
Hydrocyanic	HCN	4.9×10^{-10}		
Hydrofluoric	HF	6.8×10^{-4}		
Hydrogen chromate ion	HCrO_4^-	3.0×10^{-7}		
Hydrogen peroxide	H_2O_2	2.4×10^{-12}		
Hydrogen selenate ion	HSeO_4^-	2.2×10^{-2}		
Hydrosulfuric acid	H_2S	9.5×10^{-8}	1×10^{-19}	
Hypobromous	HBrO	2.5×10^{-9}		
Hypochlorous	HClO	3.0×10^{-8}		
Hypoiodous	HIO	2.3×10^{-11}		
Iodic	HIO_3	1.7×10^{-1}		
Lactic	$\text{HC}_3\text{H}_5\text{O}_3$	1.4×10^{-4}		
Malonic	$\text{H}_2\text{C}_3\text{H}_2\text{O}_4$	1.5×10^{-3}	2.0×10^{-6}	
Nitrous	HNO_2	4.5×10^{-4}		
Oxalic	$\text{H}_2\text{C}_2\text{O}_4$	5.9×10^{-2}	6.4×10^{-5}	
Paraperiodic	H_5IO_6	2.8×10^{-2}	5.3×10^{-9}	
Phenol	$\text{HC}_6\text{H}_5\text{O}$	1.3×10^{-10}		
Phosphoric	H_3PO_4	7.5×10^{-3}	6.2×10^{-8}	4.2×10^{-13}
Propionic	$\text{HC}_3\text{H}_5\text{O}_2$	1.3×10^{-5}		
Pyrophosphoric	$\text{H}_4\text{P}_2\text{O}_7$	3.0×10^{-2}	4.4×10^{-3}	
Selenous	H_2SeO_3	2.3×10^{-3}	5.3×10^{-9}	
Sulfuric	H_2SO_4	Strong acid	1.2×10^{-2}	
Sulfurous	H_2SO_3	1.7×10^{-2}	6.4×10^{-8}	
Tartaric	$\text{H}_2\text{C}_4\text{H}_4\text{O}_6$	1.0×10^{-3}	4.6×10^{-5}	

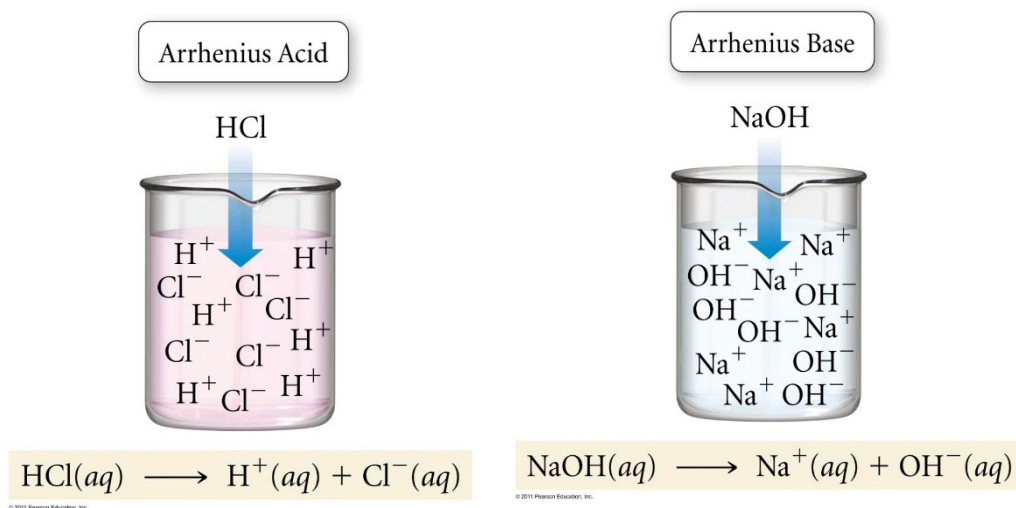
TABLE D.2 Dissociation Constants for Bases at 25°C

Name	Formula	K_b
Ammonia	NH_3	1.8×10^{-5}
Aniline	$\text{C}_6\text{H}_5\text{NH}_2$	4.3×10^{-10}
Dimethylamine	$(\text{CH}_3)_2\text{NH}$	5.4×10^{-4}
Ethylamine	$\text{C}_2\text{H}_5\text{NH}_2$	6.4×10^{-4}
Hydrazine	H_2NNH_2	1.3×10^{-6}
Hydroxylamine	HONH_2	1.1×10^{-8}
Methylamine	CH_3NH_2	4.4×10^{-4}
Pyridine	$\text{C}_5\text{H}_5\text{N}$	1.7×10^{-9}
Trimethylamine	$(\text{CH}_3)_3\text{N}$	6.4×10^{-5}



Arrhenius Theory

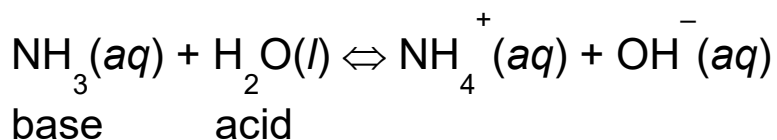
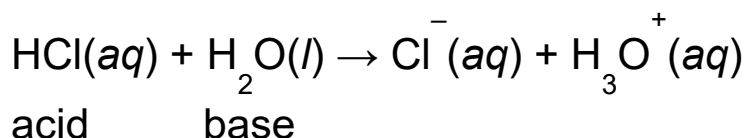
- Bases dissociate in water to produce OH^- ions and cations
- Acids ionize in water to produce H^+ ions and anions



- Does not explain why molecular substances, such as NH_3 , dissolve in water to form basic solutions – even though they do not contain OH^- ions
- Does not explain how some ionic compounds, such as Na_2CO_3 or Na_2O , dissolve in water to form basic solutions – even though they do not contain OH^- ions
- Does not explain why molecular substances, such as CO_2 , dissolve in water to form acidic solutions – even though they do not contain H^+ ions

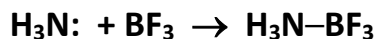
Brønsted-Lowry Theory

- In a Brønsted-Lowry acid–base reaction, an H^+ is transferred
- The acid is an H donor
- The base is an H acceptor
 - ✓ base structure must contain an atom with an unshared pair of electrons
- In a Brønsted-Lowry acid-base reaction, the acid molecule gives an H^+ to the base molecule



Lewis Acid–Base Theory

- Lewis Acid–Base theory focuses on transferring an electron pair
- Does NOT require H atoms
- The electron donor is called the Lewis Base
- The electron acceptor is called the Lewis Acid



Strong Acids

HCl

HBr

HI

HNO₃

HClO₄ *perchloric*

*H₂SO₄ *only 1st H⁺*

HBrO₃ *bromic*

HClO₃ *chloric*

Strong Bases (Metal w/ OH⁻)

LiOH

NaOH

KOH

Sr(OH)₂

Ca(OH)₂

Ba(OH)₂

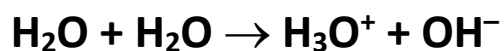
NH₄OH



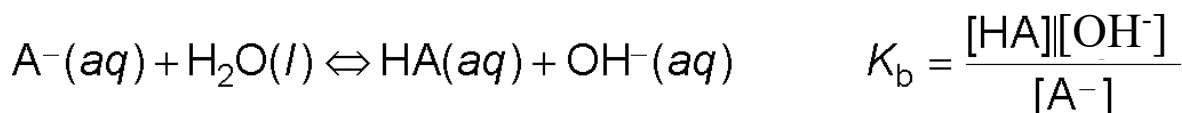
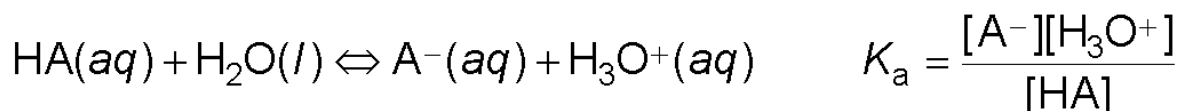
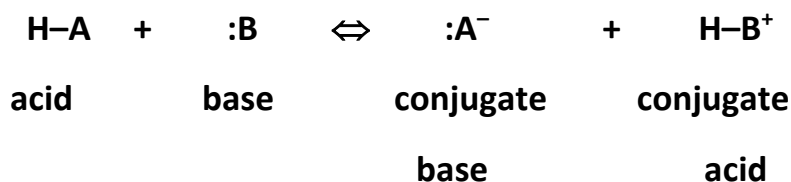
Weak Acids partially ionize (establish equilibrium)

Weak Bases partially dissociate (establish equilibrium)

Autoionization of Water



$$K_w = [\text{H}_3\text{O}^+] \times [\text{OH}^-] = 1.00 \times 10^{-14} \text{ @ } 25^\circ\text{C}$$



$$K_a \times K_b = \frac{[\text{A}^-][\text{H}_3\text{O}^+]}{[\text{HA}]} \times \frac{[\text{HA}][\text{OH}^-]}{[\text{A}^-]}$$

$$K_a \times K_b = [\text{H}_3\text{O}^+][\text{OH}^-] = K_w$$

$$\text{p}K_a + \text{p}K_b = 14.00$$

- $\text{pH} = -\log[\text{H}_3\text{O}^+] \quad [\text{H}_3\text{O}^+] = 10^{-\text{pH}}$
- $\text{pOH} = -\log[\text{OH}^-] \quad [\text{OH}^-] = 10^{-\text{pOH}}$
- $\text{p}K_a = -\log(K_a), \quad K_a = 10^{-\text{p}K_a}$
- $\text{p}K_b = -\log(K_b), \quad K_b = 10^{-\text{p}K_b}$

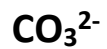
$$[\text{H}_3\text{O}^+][\text{OH}^-] = K_w = 1.0 \times 10^{-14}$$

$$(-\log([\text{H}_3\text{O}^+][\text{OH}^-])) = -\log(1.0 \times 10^{-14})$$

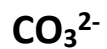
$$(-\log([\text{H}_3\text{O}^+])) + (-\log([\text{OH}^-])) = 14.00$$

$$\text{pH} + \text{pOH} = 14.00$$

1. Write the conjugate acid of the following



2. Write the conjugate base of the following.



3. Calculate the pH of 0.200 M $\text{HNO}_2(aq)$ solution @ 25 °C.

4. Calculate the pH of 0.100 M NH_3 (aq) solution @ 25 °C.

5. Calculate the pH of 0.100 M HClO_2 (aq) solution @ 25 °C.

CHE 310 – Analytical Chemistry

Quadratic Equation for Weak Acid Solutions

Use the equation below for weak acids (mono- or polyfunctional) when the exact solution to a weak acid equilibrium expression is desired (or required):

$$[\text{H}^+] = \frac{-K_a + \sqrt{(K_a)^2 + (4 \times K_a \times M_{\text{HA}})}}{2}$$

Where M_{HA} is the molarity of the component with the most acidic protons.

Quadratic Equation for Buffers

For those occasions when the Henderson-Hasselbalch equation just won't do 😊

Calculating the hydrogen ion concentration of a buffer solution using the modified quadratic equation below gives the exact solution:

$$[\text{H}^+] = \frac{-(K_a + M_{\text{A}^-}) + \sqrt{(K_a + M_{\text{A}^-})^2 + (4 \times K_a \times M_{\text{HA}})}}{2}$$

Where M_{A^-} is the molarity of the component with fewer acidic protons and M_{HA} is the molarity of the component with more acidic protons.

Programing Quadratic Equation into Calculator

Program → New → Create New →
Name → QUAD →
Program → I/O → Prompt → K,M →
 $((-K + \sqrt{K^2 + (4 * K * M)})) / 2 \rightarrow \text{Sto} \rightarrow \text{H} \rightarrow$
Program → I/O → Disp → H

The Resulting Program should look like

:Prompt K,M
: $((-K + \sqrt{K^2 + (4 * K * M)})) / 2 \rightarrow \text{H}$
: Disp H

Note: K = equilibrium constant M = Where M_{HA} and is the molarity of the component with the most acidic protons.

Programing Modified Quadratic Equation For Buffers

Program → New → Create New →
Name → BUFQUAD →
Program → I/O → Prompt → K,N,D →
 $((-(K+N) + \sqrt{(K+N)^2 + (4 * K * D)})) / 2 \rightarrow \text{Sto} \rightarrow \text{H} \rightarrow$
Program → I/O → Disp → H

The Resulting Program should look like

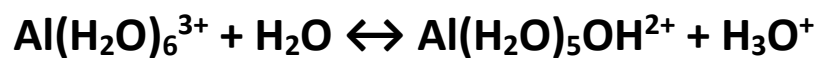
:Prompt K,N,D
: $((-(K+N) + \sqrt{(K+N)^2 + (4 * K * D)})) / 2 \rightarrow \text{H}$
: Disp H

Notes: K = equilibrium constant, N = M_{A^-} and is the molarity of the component with fewer acidic protons, and D = M_{HA} and is the molarity of the component with more acidic protons.

Salt Hydrolysis

Reaction of salt with water (causes water to split)

1. Group I and II cations never hydrolyze
2. Conjugate Bases of strong acids never hydrolyze
3. Conjugates of weak acids/bases do hydrolyze
4. Transition metals ionize and act as weak acids



6. Calculate the pH of 0.100 M $\text{NaCHO}_2(aq)$ solution @ 25 °C.

7. Calculate the pH of 0.200 M NH_4Cl (aq) solution @ 25 °C.

Buffer is a mixture of a weak acid/base with its conjugate

Acetic acid/sodium acetate

$\text{NH}_3/\text{NH}_4\text{Cl}$

Buffer is a solution that tends to resist changes in pH either through dilution or addition of acids or bases.

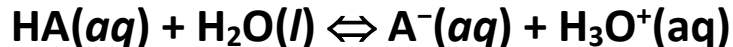
Buffer Capacity depends on quantity of weak acid/base and the conjugate.

Making Buffers

Mix weak acid/base and conjugate (acetic acid and sodium acetate)

Add strong base to a weak acid (converts some of the acid to the conjugate)

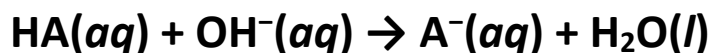
How Acid Buffers Work:



Addition of Acid



Addition of Base



Adding salt that contains the common ion



to the left

$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$

$$K_a = [\text{H}^+] \times \frac{[\text{A}^-]}{[\text{HA}]}$$

$$pK_a = pH - \log \frac{[\text{A}^-]}{[\text{HA}]}$$

$$pH = pK_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

$$\text{pH} = \text{p}K_a + \log \frac{[\text{conjugate base anion}]_{\text{initial}}}{[\text{weak acid}]_{\text{initial}}}$$

For most problems, this means that the initial acid and salt concentrations should be over 100 to 1000x larger than the value of K_a

Fails when K_a is $> 10^{-3}$

Fails when $\text{p}K_a$ is < 3

Fails when x is not small

8. What is the pH of a buffer that is 0.050 M $\text{HC}_7\text{H}_5\text{O}_2$ and 0.150 M $\text{NaC}_7\text{H}_5\text{O}_2$.

K_a for $\text{HC}_7\text{H}_5\text{O}_2 = 6.5 \times 10^{-5}$

Titration of a Weak Acid with a Strong Base

- 1) The initial pH is determined using the K_a of the weak acid
 - a) The initial pH is that of the weak acid solution
 - b) calculate like a weak acid equilibrium problem
- 2) The pH in the excess acid region is determined as you would determine the pH of a buffer
 - a) Before the equivalence point, the solution becomes a buffer
 - b) calculate moles HA_{init} and moles A^-_{init} using reaction stoichiometry
 - c) calculate pH with Henderson-Hasselbalch using moles HA_{init} and moles A^-_{init}
 - d) Half-neutralization $pH = pK_a$
- 3) The pH at the equivalence point is determined using the K_b of the conjugate base of the weak acid
 - a) At the equivalence point the solution has only the conjugate base anion in it before equilibrium is established
 - b) moles A^- = original moles HA
 - i) calculate the volume of added base
 - c) $[A^-]_{init} = \text{moles } A^- / \text{total volume}$
 - d) calculate like a weak base equilibrium problem
- 4) The pH after equivalence is dominated by the excess strong base
 - a) the basicity from the conjugate base anion is negligible
 - b) Beyond equivalence point, the OH^- is in excess
 - c) $[OH^-] = (\text{moles } OH^- \text{ excess}) / \text{total volume}$
 - d) $[H_3O^+][OH^-] = 1 \times 10^{-14}$

A 0.4456 g sample of KHP is dissolved in 50.00 mL of water and titrated with 0.09910 M NaOH.

K_a for KHP = 3.89×10^{-6}

9a) Calculate the initial pH.

9b) Calculate the pH after adding 10.00 mL of NaOH (0.09910 M)

9c) Calculate the pH at the equivalence point.